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Heterogeneity in passenger satisfaction of bus rapid transit system among age and gender groups: A PLS-SEM Multi-group analysis

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ABSTRACT

Public transport is a promising way to solve urban mobility problems such as traffic congestion and vehicular emissions. However, in order to encourage commuters to use the public transport facility, operators need to provide a high level of service to ensure their satisfaction, which is one of the crucial factors influencing the mode choice of commuters. Despite this, there is limited understanding of the factors influencing passenger satisfaction in public transportation, especially in developing countries like India. To address this gap in knowledge, the present study explores the factors affecting passenger satisfaction on bus rapid transit system (BRTS), and examines the differences in the perception of satisfaction for passengers of different age and gender groups. The present study utilizes Principal component analysis (PCA) and Partial Least Square Structural Equation Modelling (PLS-SEM) to explore passengers' satisfaction based on 26 key attributes of the BRTS. PCA uncovers six dimensions explaining the overall satisfaction of passengers, i.e., *connectivity, comfort & convenience, safety, cleanliness, station facilities, and operations* and the results of PLS-SEM reveal that all these dimensions positively influence passenger satisfaction. The results of multigroup analysis reveal that there exist heterogeneity in the perception of satisfaction for passengers of different gender and age groups. Unlike female passengers, who perceived *connectivity* and *comfort & convenience* higher, male passengers perceived *operations* and *station facilities* higher. Furthermore, it was found that passengers with age $\geq$ 40 perceived *station facilities* and *cleanliness* higher than passengers with age <40. Taking into account the heterogeneity among passengers of different groups. the study suggests various measures to ensure passenger satisfaction on the BRTS based on Importance Performance Map analysis (IPMA).

1. Introduction

Around the world, public transport plays a crucial role in providing quality transport facilities to different sectors of society. It provides citizens with an adequate transport facility to travel to their workplaces, educational institute, and recreational areas. Unlike personal vehicles, public transport allows more road users to travel together on a designated route. Thereby establishing itself as a promising way to reduce traffic congestion and vehicular emissions (Bull, 2004; Holmgren, 2007; Rajé et al., 2018; Titos et al., 2015; Wirasinghe et al., 2013). In developing countries like India, the rapid population growth in cities due to unplanned urbanization and an increasing trend of private vehicle ownership have increased travel demands and traffic congestion (Imre and Celebi, 2017; Li et al., 2019). The increasing demand of Indian commuters has resulted in considering public transport alternatives such as rapid transit metro, light rail transit (LRT), and bus rapid transit (BRT) corridors (Deng and Nelson, 2011). Because of the higher investment required to set up RTM and LRT, BRT is becoming popular nowadays because of lower initial investment (Deng and Nelson, 2011). The Institute of Transportation & Development policy defines BRT as "*a high-quality bus-based transit system that delivers fast, comfortable, and cost-effective urban mobility through the provision of segregated right-of-way infrastructure, rapid and frequent operations, and excellence in marketing and customer service*" (Institute for Transportation & Development Policy (ITDP), 2016). A well-planned BRTS exhibits a potential benefit in increasing the capacity, accessibility, reliability, safety and decreasing travel time.

One of the major aims of implementing BRTS is to meet the mobility demand of commuters and encourage the mode shift by providing quality transport service. It is evident that better service quality results in increased usage of public transport facilities, which further reduces traffic congestion and environmental emissions (De Oña et al., 2013; Li

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et al., 2019). Previous studies have pointed out that the users of a public transport system are the sole judges of the service quality (Aydin, 2017; De Oña et al., 2013; TCQSM, 2013). Therefore, in recent years, many researchers have focussed on assessing passenger satisfaction to measure the perceived service quality of the system (Eboli and Mazzulla, 2012; Li et al., 2019; Redman et al., 2013; Shang et al., 2019; Wan et al., 2016). Passenger satisfaction is defined as *"the sense of either enjoyment or displeasure, which results from a comparison between the passenger's expectation from the service and the real service provided"*. Passenger satisfaction is an essential factor that measures the quality of a public transport facility and is crucial in deciding commuters' mode choice. The lack of passenger satisfaction may cause a mode shift of passengers from public transport to private cars (Redman et al., 2013). Therefore, it is essential for authorities to improve the service quality of the public transport system to encourage the usage of the service (Allen et al., 2019a).

In India, the first implementation of BRTS was done in the year 2006, and since then many BRTS have been implemented in India. Combined, all these systems carry a huge number of passengers i.e. around 4,97,411 passengers/day (BRTdata.org). However, transit agencies have paid very little attention on investigating passengers' satisfaction on Indian BRT systems. Further, previous studies suggest that passengers of different socioeconomic groups perceive public transport satisfaction distinctly. Yet, very little attention has been paid by transit agencies in exploring these effects. Thus, it is imperative to explore passengers' satisfaction on the BRT system and observe the heterogeneity among passengers of different socioeconomic groups to enhance passengers' satisfaction on the BRT system. The present study examines the less-explored gender and age aspects in bus transit satisfaction evaluation by highlighting the disparity in perception of satisfaction for passengers of different ages and gender. Prior, studies also highlights the need for gender and age-segregated studies on transit satisfaction evaluations realizing the differences in perceptions, expectations, and challenges faced by individual groups of passengers (ITF, 2018; Mandhani et al., 2021). To the best of authors' knowledge, this study is a novel attempt to examine the determinants of passenger satisfaction on BRTS among different groups of gender and ages. The results of the present study will motivate transit agencies in developing nations like India to integrate gender and age effects in ongoing as well as upcoming transit projects and work out optimum strategies to meet the demand of passengers belonging to different age groups and gender. Further, the present study identifies gender and age-specific interventions to enhance the overall satisfaction on BRTS using importance-performance map analysis (IPMA). This analysis act as a prescription tool for transit agencies in formulating gender and age-specific improvement strategies. Moreover, such an extensive study on satisfaction evaluation on BRTS has been carried out for the first time in a developing country like India, which is geographically and culturally very different from studies available in the literature.

The present study has been structured into six sections. Section 2 presents a review of existing literature on passenger satisfaction. Section 3 describes the methodological framework adopted in the present study and the results are presented in Section 4. Section 5 provides a detailed discussion of the obtained results, and Section 6 presents the policy implications drawn from the study. Finally, conclusion and limitations are presented in Section 7.

2. Previous studies on passenger satisfaction

In the past, many researchers have focussed on exploring the relationship between service quality factors and passenger satisfaction for commuters using the public transport service. To measure the service quality of the public transport system, satisfaction surveys are usually carried out, in which the users are asked to rate various attributes of public transport based on their perception. These responses are collected either via online surveys (such as Google form or online surveys), on-board interviews (i.e., in-personal interviews of users), or by telephonic interviews (Li et al., 2020). Previous studies reveal that there exists a strong relationship between passenger satisfaction and service quality. Therefore, many researchers have focussed on assessing passenger satisfaction on various public transport modes to improve the overall performance of the system (Allen et al., 2019a; Aydin, 2017; De Oña et al., 2013; Redman et al., 2013; Tyrinopoulos and Antoniou, 2008; Wan et al., 2016). Various methods such as factor analysis, regression analysis, structural equation modeling (SEM), clustering, etc., have been used to study passenger satisfaction on different public transport facilities and to determine the factors affecting passenger satisfaction. Parasuraman et al. (1988) developed a 22-item instrument (also known as SERVQUAL) to assess the passenger perception of service quality. Parasuraman et al. (1988) proposed five dimensions affecting service quality, i.e., tangibles, reliability, responsiveness, assurance, and empathy. De Oña et al. (2013) used Structural equation modeling (SEM) to evaluate the service quality of the bus transit service based on users' perceptions. The analysis revealed three aspects of passenger satisfaction defining the overall service quality of the transit system. These were service (included attributes related to the performance of the service, i.e., frequency, speed, fare, etc.), comfort (included attributes liked to comfort experienced by the user, i.e., cleanliness, temperature, accessibly, etc.), and personnel (included attributes related to the behavior of staff). Allen et al. (2019a) performed PCA and SEM analysis to evaluate passenger satisfaction for BRTS in Latin America. PCA revealed four components of passenger satisfaction, namely reliability, safety, comfort, and customer service. Based on the analysis, Allen et al. (2019a) confirmed the existence of a hierarchy of transit needs (i.e., pyramid of transit needs) where reliability was at the bottom of the pyramid, followed by safety, customer services, and comfort. Tyrinopoulos and Antoniou (2008) analyzed the level of satisfaction among the users of 5 different public transit systems operating in three different modes (bus, trolley bus, and rail/metro) in Greece using PCA and ordered logit model. PCA revealed four factors: quality of service, transfer quality, service production, and information/courtesy. Further, ordered logit models revealed that the most important satisfaction attribute varied across the systems. For the metro system, vehicle cleanliness, the behavior of staff, and ticketing systems were the three most important attributes whereas, for buses and trolley bus operators, the most important satisfaction attributes were frequency, vehicle cleanliness, and network coverage. In New York, Wan et al. (2016) investigated the passenger's perception of satisfaction on the New York Select bus service BRTS, using statistical analysis and regression modeling. The study revealed that service frequency, speed, and on-time performance were significantly associated with rider satisfaction on New York Select bus service. Eboli and Mazzulla (2007) performed factor analysis and SEM to explore the relationship between service quality attributes and customer satisfaction for the bus transit system of Cosenza, Italy. It was found that service planning and reliability had a major effect on customer satisfaction. De Oña et al. (2016) performed cluster analysis to study the service quality of buses in Granada, Spain. To identify the vital factors affecting the service quality of system, decision tree methodology was used in the study. The results revealed that frequency, speed, punctuality, space, and safety were the most important factors affecting the system's service quality. In the previous studies researchers have proposed various attributes to define the service quality of public transport. Redman et al. (2013) categorized these quality attributes into two categories; physical attributes and perceived attributes. Physical attributes require the field data collected by the agencies to measure the service quality of a Public transport system, such as speed, frequency, price, etc. Whereas, perceived attributes requires responses from the users regarding their perception of service quality (Redman et al., 2013; Wan et al., 2016).

In the past, various researchers have pointed out that passenger satisfaction varies with the socioeconomic characteristics of the passengers (Börjesson and Rubensson, 2019; Mandhani et al., 2021;

Nguyen-Phuoc et al., 2021). Some of the recent studies (Mandhani et al., 2021; Yuan et al., 2021b) have also highlighted the need to consider the unobserved heterogeneity among passengers belonging to different groups. However, knowledge about the existence of heterogeneity in the perception of satisfaction for passengers of different age groups and gender, specifically in the case of bus-based transit systems is still very limited. Considering the enormous number of passengers traveling on Indian BRT buses, it is important to investigate the determinants of passenger satisfaction on the BRT system and also, to understand the disparity in perception of satisfaction for passengers of different ages and gender. Therefore, the present study aims to i) investigate various attributes of BRTS that affect passengers' satisfaction. ii) explore the differences in perceived satisfaction for passengers of different age groups and gender, and iii) highlight various gender and age-specific countermeasures/policies to improve passengers' satisfaction.

3. Methodology

3.1. Questionnaire survey

To collect the data from the passengers, a questionnaire survey was designed in the present study, consisting of four major sections, aimed to collect passenger's demographic information, trip information, passenger's satisfaction with various service attributes of BRTS, and passenger's overall satisfaction. In the first section, the passengers were asked to provide their demographic information such as age, gender, level of education, and occupation. The second section asked the passengers to respond about their trip characteristics, such as travel frequency and trip purpose. In the third section, passengers were asked to respond to their level of satisfaction toward the BRT service based on 26 key attributes of passenger satisfaction. The selection of key attributes for the study was based on the review of existing literature on passenger satisfaction. During the survey, the passenger was asked to rate the attributes based on their perceived satisfaction on a 6-point Likert scale; where 1 being the lowest score (dis-satisfied) and 6 being the highest (highly satisfied). Finally, the passengers were asked to evaluate their overall satisfaction with the BRTS on a 6-point Likert scale. After developing the questionnaire, the survey was sent to an expert panel consisting of three transport researchers having experience of more than ten years. A pre-test was also performed on 50 known group of passengers of Ahmedabad BRTS, and based on their recommendation, some of the problematic items were modified. Finally, the survey form was floated via emails and online social media platforms.

3.2. Study design and procedure

In the present study, the data was collected for the Ahmedabad BRTS, which is one of the longest and successful implementations of BRTS in India. The system has a total network length of around 84.5 km, with an average ridership of 1,30,000 passengers/day (BRTdata.org). In the present study, passengers' responses on the questionnaire survey were collected through an online survey form during June–November 2021. Apart from sharing the survey form/link via social media platforms, the survey form was also floated within government offices and private offices and educational institutes lying within the vicinity of Ahmedabad BRTS. To get a higher response on the questionnaire survey, we followed the snowball sampling technique, in which the participants were asked to refer the survey form to other potential participants. The survey form was kept entirely anonymous (i.e., without asking participants' name, email id, and contact number), and the participants were assured that the collected data would be used only for research purposes. In total, we received 812 responses, out of which 102 participants responded that they were not frequent users of the BRTS, i.e., having a frequency of using the service less than twice a month. For the present study, we only focused on the frequent users of the service (n = 710). To check whether our data followed the assumption of multivariate normality, one-sample Kolmogorov-Smirnov test (K–S test) was performed on the collected data using SPSS software. The test revealed that the K–S statistics values were in the range of 0.133–0.240 with a p-value of <0.001, indicating the data did not conform to normal distribution. Therefore, PLS-SEM was used in the present study as it does not require the data to follow the assumption of multivariate normality (Hair et al., 2017; Shen et al., 2016).

3.3. Sample size determination

Determining adequate sample size is essential to ensure the quality of results. As a rule of thumb, it is suggested that the minimum sample size is ten times the maximum number of structural paths pointing at a latent variable (Hair et al., 2017). However, the present study determined the sample size using more rigorous criteria given by the following statistical formula (Johnson and Wichern, 2007; Tyrinopoulos and Antoniou, 2008).

$$Sample\ Size\ (S.S) = n\left[1 + \frac{(n-1)}{p(1-p)}\left(\frac{d}{Z_{\alpha/2}}\right)^2\right]^{-1} \quad (1)$$

Here, n is the daily average ridership of Ahmedabad BRTS, which is 1,30,000 passengers per day (BRTdata.org); p is the probability of selecting a choice (satisfaction), for a neutral situation $p = 0.5$ (Tyrinopoulos and Antoniou, 2008); d is the confidence level or margin of error ($d = 0.05$) and $Z_{\alpha/2} = 1.96$ for a confidence level of 95%. Based on **Equation (1)**, the minimum sample size for the present study was found to be 313. Further, to ensure the sample size was adequate enough to give logical findings, a post-hoc power analysis test was performed using the G-Power 3.1 software (Erdfelder et al., 2009). For a sample size of 710 and an effect size of f = 0.3 (moderate effect), the statistical power of the study was $\geq$0.99 for a confidence level of 95%. Statistical power $\geq$0.99 confirms the adequacy of the sample size collected in the present study (Bivina and Parida, 2019; Cohen, 1988).

3.4. Data analysis

Initially, descriptive statistics were performed on the data to analyze the participants' demographic characteristics and their satisfaction rating on 26 key attributes of BRT service. For the further analysis of survey data, two statistical methods were used in the study, i.e., Principal component analysis and Partial least square - Structural equation modeling (PLS-SEM). The results from these methods were used to assess the passenger perceived satisfaction with respect to BRTS of Ahmedabad. To investigate the differences in perceived satisfaction among the passengers of different groups, a Multigroup analysis was performed in the study. Finally, an Importance performance map analysis (IPMA) was performed to recommend suggestions for ensuring passenger satisfaction. Fig. 1 presents the analytical steps performed in the present study.

4. Results

4.1. Descriptive statistics

Fig. 2 provides detailed descriptive statistics of passengers' participated in the study. Out of 710 respondents, 51.4% of the respondents were female passengers, whereas 48.6% were male passengers. The majority of the respondents were in the age group of 20–29 years. However, very few participants were aged above 50 years. Regarding the purpose of the trip, most of the passengers (47.5%) responded taking the BRT service for their work trips. Further, 37% of the respondents mentioned that they were commuting for educational purposes such as colleges and school, whereas 15.5% of the respondents were traveling for recreational purposes. Among all the participants, the majority of them responded that they were traveling almost daily, and 30.3% were traveling at least 3 times in a week. Interestingly, the statistics reveal

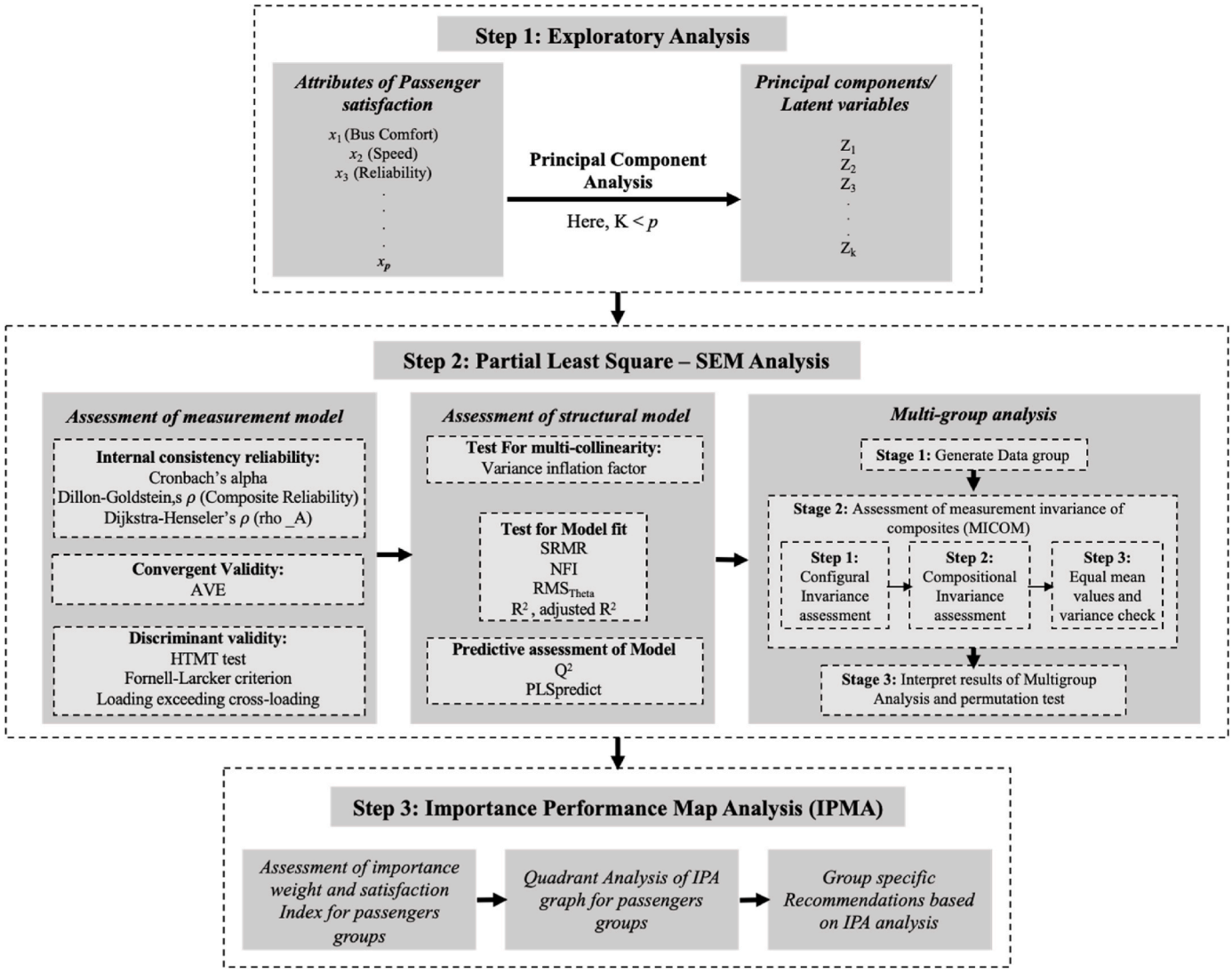


Fig. 1. Methodological framework adopted in the study.

that 63.1% of the passengers have their own personal vehicle; however, they prefer to travel via BRTS.

Based on passengers' responses on the survey, Fig. 3 reveals the descriptive statistics of passengers' perception of satisfaction towards individual attributes and overall satisfaction on the Ahmedabad BRTS. The higher mean score on frequency [M = 4.65, S.D = 1.15] followed by safety against accident [M = 4.50, S.D = 1.15] reveals that most of the passengers were moderately satisfied with the frequency of buses in the system and passengers feel safe against road accidents while taking the service. Further, the majority of the passengers [M = 4.46, S.D = 1.08] were moderately satisfied with the attitude and behavior of BRTS personnel's when communicating or having transactions with them. The lowest mean score on facilities for persons with disability (PWD) [M = 2.98, S.D = 1.28] and ease of boarding and alighting for PWD [M = 2.26, S.D = 1.14] reveals that the system fails to provide the required services for the disabled commuters. Regarding connectivity, around 25% and 24% of the passengers responded highest on the Likert scale for multimodal and local connectivity, respectively, signifying that the system has good connectivity with most of the areas within the city and paratransit services are available at the BRTS stations. However, for park & ride availability, around 55% of the passengers rated less than equal to 3, indicating the lack of parking facilities at/or near the stations. In safety, passengers perceived the safety against theft [M = 3.99, S.D = 1.17] and road accidents [M = 4.50, S.D = 1.15] better than women safety [M = 3.49, S.D = 1.29]. Further, on the overall satisfaction scale, around 79% of the passengers rated ≥4, indicating that most passengers were moderately satisfied with the BRT service.

4.2. Principal component analysis (PCA)

As an initial step, Principal component analysis with varimax rotation was performed on the passenger responses using SPSS statistics software to uncover if there exists a relationship between various attributes of passenger satisfaction and which of these attributes can form a construct as a whole. In line with previous studies, only items with a factor loading >0.5 were considered (Bivina et al., 2019). PCA revealed six components having an Eigenvalue greater than 1. A KMO (Kaiser-Meyer-Olkin) measure of sampling adequacy = 0.929 and a statistically significant BTS (Bartlett's test of sphericity) value of ($\chi^2(231) = 1189.78$, $p < 0.001$), indicated that the data was suitable for the analysis. Overall, the six-component solution explained 66.15% of the total variance; *connectivity, comfort & convenience, safety, cleanliness, station facilities and operations* (Table 1).

4.3. Partial least square – SEM

Once the latent variables (constructs) were found using the principal component analysis, a conceptual framework for passenger satisfaction

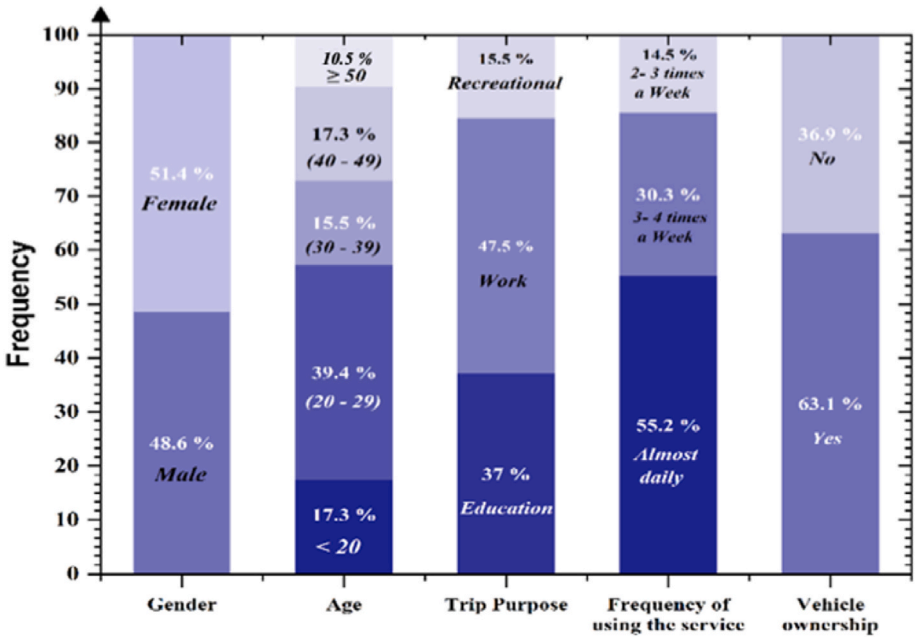


Fig. 2. Descriptive statistics of the participants in the study.

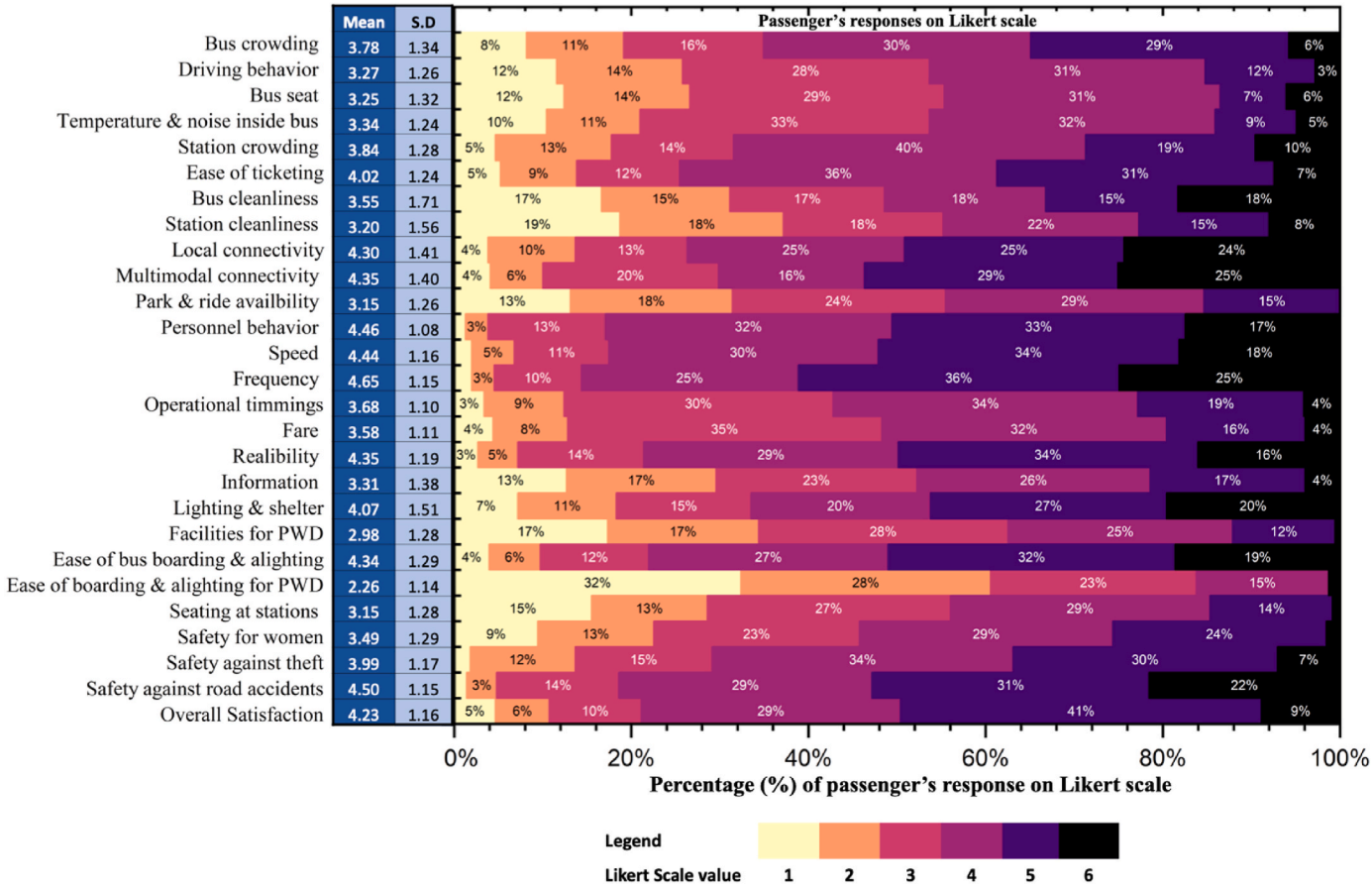


Fig. 3. Descriptive statistics of passenger's responses on the questionnaire.

was created as shown in Fig. 4. In the framework, exogenous latent variables are depicted by solid ovals, observed variables or manifested variables are depicted by a solid rectangle, and a solid grey oval depicts endogenous latent variables. As recommended by Hair et al. (2017), first measurement model was assessed to examine the relationship between the manifested variables and the latent variables (constructs) and to check if the measurement model was valid and reliable by performing validity and reliability test on the measurement model. Once the

Table 1
Results of Principal component analysis.

Attribute/Items	Component					
	Comfort & Convenience	Station facilities	Operations	Connectivity	Safety	Cleanliness
Bus Crowding (CC1)	0.767					
Temperature & noise inside bus (CC2)	0.717					
Driving behavior (CC3)	0.750					
Station crowding (CC4)	0.795					
Ticket counter (CC5)	0.733					
Bus seat (CC6)	0.689					
Bus cleanliness (C1)						0.869
Station cleanliness (C2)						0.886
Local connectivity (CON1)				0.758		
Multimodal connectivity (CON2)				0.784		
Park & ride availability (CON3)				0.800		
Reliability (OP1)			0.686			
Speed (OP2)			0.692			
Frequency (OP3)			0.663			
Fare (OP4)			0.762			
Personnel behavior (OP5)			0.696			
Service operational timing's (OP6)			0.805			
Facilities for PWD (tactile paver, ramps, level boarding etc.) (SF1)		0.698				
Information (SF2)		0.720				
Lighting & shelter at stations (SF3)		0.641				
Ease of boarding & alighting from bus (SF4)		0.707				
Ease of boarding & alighting for PWD (SF5)		0.772				
Seating at stations (SF6)		0.727				
Safety against accident (S1)					0.745	
Safety against theft (S2)					0.800	
Safety for women (S3)					0.731	
No. of items	6	6	6	3	3	2
Mean	3.585	3.352	4.194	3.933	3.994	3.374

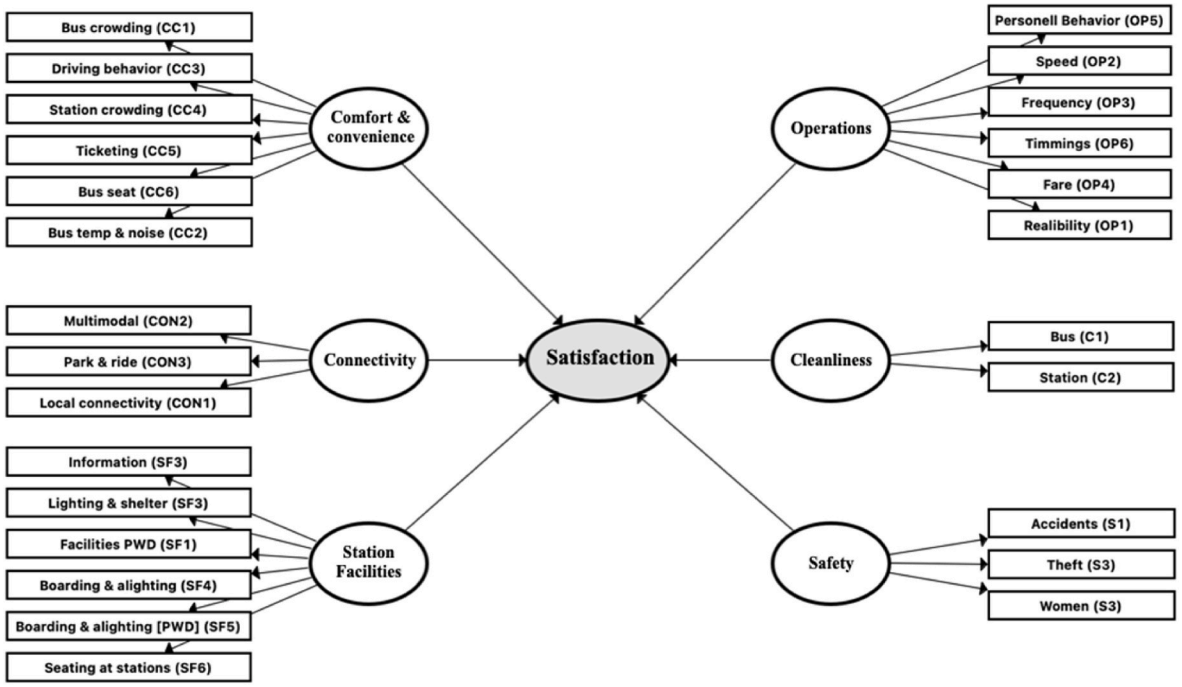


Fig. 4. Conceptual framework for passenger satisfaction.

measurement model was found to be valid and reliable, the structural model was evaluated to investigate the relationship between the latent variables and passenger satisfaction.

As the survey data was self-reported, it is essential to determine the presence of common-method variance bias among the observed variables as there is a possibility of common method bias, including consistency motif and social desirability. Common method bias occurs when the part of common variance shared among the construct is attributed to common measurement and the same source (Kaltsonoudi et al., 2021; Podsakoff et al., 2003). In this case, same error source will appear among all variables. In order to determine the presence of common method variance bias in the data, the Harman one-factor test was performed on the variables (Farooq et al., 2018). For this, principal component analysis (without rotation) was performed on all the variables with only one-factor solution. The results reveal that the single factor accounts for the variance less than 50%, thereby indicating that the data does not

have any common method variance bias issue (Kaltsonoudi et al., 2021).

4.3.1. Assessment of measurement model

Initially, confirmatory factor analysis was performed to confirm and refine the items and latent variables in the measurement model. For the assessment of the measurement model, Henseler et al. (2016a) suggested three criteria to test the measurement model: internal consistency reliability, convergent validity, and discriminant validity.

4.3.1.1. Internal consistency reliability. To test the internal consistency among observed variables within each construct (latent variable), the Cronbach's alpha value was estimated for each construct. As depicted in Table 2, Cronbach's alpha values ranged between 0.795 and 0.874, which is higher than the threshold value of 0.70 (Henseler et al., 2016a). Further, for each construct, the composite reliability (Dillon-Goldstein's ρ) and rho _A (Dijkstra-Henseler's ρ) values were greater than 0.88 and 0.80, respectively, which satisfies the minimum criteria of 0.70 (Henseler et al., 2016a). The results indicate that there exists good internal consistency reliability among the constructs.

4.3.1.2. Convergent validity. For the measurement model, the convergent validity was assessed by evaluating average variance extracted (AVE) and based on the factor loading of each observed variable. As depicted in Table 2, all the observed variables had significant ($p < 0.001$, t-value >2.57) acceptable factor loading in the range of 0.731 and 0.917 with their respective construct, indicating good reliability of the observed variable. Further, for all the latent variables, the average variance extracted (AVE) value was higher than 0.60, which is higher than the threshold value of 0.50 (Henseler et al., 2016a). Therefore, the results indicate that the convergent validity was adequate.

4.3.1.3. Discriminant validity. For the assessment of discriminant validity, Fornell-Larcker criteria was used to evaluate the measurement model. The square root of the AVE values for all the constructs were greater than the corresponding square correlation with other constructs, thus indicating the discriminant validity of the constructs (Table A1).

Table 3
Path relationship in the structural model.

Path Relationship	Path coefficient (ß)	S.D	t-value	p-value
Cleanliness - > Satisfaction	0.144	0.022	6.502	<0.001
Comfort & Convenience - > Satisfaction	0.250	0.032	7.748	<0.001
Connectivity - > Satisfaction	0.251	0.027	9.457	<0.001
Operations - > Satisfaction	0.267	0.028	9.459	<0.001
Safety - > Satisfaction	0.126	0.026	4.855	<0.001
Station Facilities - > Satisfaction	0.183	0.030	6.066	<0.001

Moreover, as recommended by Henseler et al. (2016a), the Heterotrait-monotrait ratio of correlation (HTMT) values in the present study were found well below 0.726 (Table A2), which is less than the threshold value of 1 (Table A2). Further, cross-loading of indicators was also checked to evaluate the discriminant validity of the constructs. For a measurement model to fulfill discriminant validity criteria, it is expected that all the observed variables should have the highest loading in their respective construct only. It was observed that all the observed variables had the highest loading on their respective construct (Table A3). Henceforth, the measurement model demonstrates excellent discriminant validity based on the three tests.

4.3.2. Structural model evaluation

Once the measurement model was evaluated for appropriateness, the next step was to evaluate the structural model. In PLS-SEM, the assessment of the structural model is done on the basis of parameters such as coefficient of determination (R^2), standardized root mean square residual (SRMR), RMS_{theta}(Root mean square residual covariance), Normalized fit index (NFI) and predictive relevance (Q^2). Before assessing the structural model, the VIF (variance inflation factor) values for all the indicators and latent variables were examined to ensure that there is no issue of multicollinearity. The VIF values for all the indicators/manifested variables were between 1.646 and 2.034, whereas for the latent variable, the VIF values were in the range of 1.155 and 1.777 (Table B1). For both latent variables and indicators, the VIF value

Table 2
Construct reliability and Validity of measurement model.

Latent variable	Observed variable	Loading	T Statistics	Average variance extracted	Dillon-Goldstein,s ρ (Composite Reliability)	Cronbach's α	Dijkstra-henseler's ρ (rho _A)
Comfort & Convenience	CC1 (Bus crowding)	0.766	37.229*	0.613	0.905	0.874	0.881
	CC2 (Bus temp & noise)	0.822	57.048*				
	CC3 (Driving behavior)	0.799	47.671*				
	CC4 (Station crowding)	0.796	48.024*				
	CC5 (Ticketing)	0.746	29.417*				
	CC6 (Bus seat)	0.769	37.974*				
Station Facilities	SF1 (Facilities PWD)	0.754	35.134*	0.604	0.902	0.869	0.871
	SF2 (Information)	0.777	40.692*				
	SF3 (Lighting & shelter)	0.747	31.627*				
	SF4 (Boarding & alighting)	0.799	46.609*				
	SF5 (Boarding & alighting for PWD)	0.776	40.408*				
	SF6 (Seating at stations)	0.810	45.532*				
Operations	OP1 (Reliability)	0.731	26.125*	0.601	0.900	0.867	0.870
	OP2 (Speed)	0.786	34.526*				
	OP3 (Frequency)	0.803	43.273*				
	OP4 (Fare)	0.794	40.907*				
	OP5 (Personnel Behavior)	0.744	26.272*				
	OP6 (Operational timings)	0.791	39.788*				
Connectivity	CON1 (Local connectivity)	0.889	84.897*	0.739	0.894	0.823	0.830
	CON2 (Multimodal)	0.832	47.554*				
	CON3 (Park & ride)	0.856	60.859*				
Safety	S1 (Accidents)	0.839	47.180*	0.714	0.882	0.800	0.800
	S2 (Theft)	0.854	55.102*				
	S3 (Women)	0.842	56.395*				
Cleanliness	C1 (Bus)	0.917	88.988*	0.830	0.907	0.795	0.797
	C2 (Station)	0.905	71.141*				

*$p < 0.001$.

did not exceed the threshold value of 5, indicating no issue of multicollinearity.

The results indicate that the proposed model has 73.2% of explanatory power for passenger satisfaction ($R^2 = 0.732$). As suggested by Henseler et al. (2016a), a value of R^2 was greater than 0.67, indicates sufficient model fit. Further, predictive relevance (Stone-Geisser's Q^2) for the satisfaction model, calculated using blindfolding with an omission distance of 7, was 0.724, which is greater than 0, thus indicating adequate predictive quality of the model. Besides, PLSpredict was also performed on the overall model to measure the model's out-of-sample predictive power. As the prediction errors were not symmetrically distributed, we compared MAE (mean absolute error) for the PLS-SEM model and Linear regression model (LM) (Galit et al., 2019). The results reveals that MAE for PLS-SEM (MAE = 0.481) was smaller than LM (MAE = 0.492). Further, the predictive relevance (Stone-Geisser's Q^2) value for PLS-SEM ($Q^2 = 0.713$) was less than LM ($Q^2 = 0.725$). Thereby suggesting high predictive power of the model (Galit et al., 2019). Moreover, the overall model fit for the passenger satisfaction model was also well within the threshold values i.e. NFI = 0.87 ≥ 0.80; SRMR = 0.051 < 0.08; RMS_{theta} = 0.118 < 0.12, thus indicating acceptable overall goodness of fit (Henseler et al., 2016a).

In the final step of structural model evaluation, bootstrapping (5000 sub-samples) was performed to check the significance and relevance of the path coefficient. Table 3 reveals that all the latent variables have a significant positive effect on the overall passenger satisfaction with a *p*-value less than 0.001. It can be inferred from the results that the passenger satisfaction is positively influenced by factors such as connectivity ($\beta = 0.251$, t = 9.457), comfort & convenience ($\beta = 0.250$, t = 7.748), operations ($\beta = 0.267$, t = 9.459), cleanliness ($\beta = 0.144$, t = 6.502), station facilities ($\beta = 0.183$, t = 6.066) and safety ($\beta = 0.126$, t = 4.855). One of the strongest predictors of passenger perceived satisfaction was "*operations*". Furthermore, "*connectivity*" and "*comfort and convenience*" was also significantly associated with the passenger perceived satisfaction. Among all the latent variables, *safety* has the lowest impact on passenger satisfaction, followed by cleanliness.

4.3.3. Multigroup analysis

In the present study, multigroup analysis was performed to investigate the differences between two groups of passengers towards satisfaction on BRTS; i.e., male-female and passengers with age ≥40 - age <40. Before performing multigroup analysis for the passenger groups, measurement invariance was determined for both the groups by performing MICOM three-step procedure (Henseler et al., 2016b; Latan and Noonan, 2017). As the first step of MICOM procedure, configural invariance was examined to check whether the same basic factor structure exists in both groups. As identical indicators, treatment, and an identical algorithm were performed on both groups of gender and age, the configural invariance was established. In the second step, compositional invariance was examined for both groups (Table C1). For both groups of age and gender, the original correlation was greater than or equal to 5% quantile; hence compositional invariance was established (Henseler et al., 2016b). Finally, the third step of MICOM procedure was performed on the passenger's groups to check if full measurement invariance was established on the basis of equal mean and equal variance (Table C2). The mean and variance original difference falls within 2.5% and 97.5% boundaries; thus, full measurement invariance was established for the groups of gender and age. Finally, the result of MICOM three-step procedure suggests that we can perform multigroup analysis on both groups.

Table 4 presents the results of the Multigroup analysis for both groups. In multigroup analysis, a p-value less than 0.05 (5% significant level) for Henseler's MGA and permutation test indicates significant differences among the groups for specific path models (Nguyen-Phuoc et al., 2021). The result of multigroup analysis for male-female passengers indicates that in terms of passenger satisfaction, there are significant differences among male and female passengers with regard to comfort & convenience, connectivity, operations, and station facilities. Female passengers perceived comfort & convenience, and connectivity higher than male passengers. Whereas, male passengers perceived operations and station facilities higher than female passengers. Further, for passengers of different age groups, it was observed that passengers with age ≥40 perceive station facilities and cleanliness higher than passengers with age <40.

4.4. Importance performance analysis

The passenger satisfaction index of indicators and dimensions along with their effect on overall satisfaction are shown in Fig. 5. The importance (derived importance) of indicators and service dimensions was based on the path coefficient with the target construct (i.e., overall

Table 4
Multigroup Analysis for passenger groups.

Group 1: Female vs Male						
Relationship	Path Coefficient (Female)	Path Coefficient (Male)	Path Coefficient difference (Female-Male)	p-value		Significant difference?
				Henseler's MGA	Permutation test	
Cleanliness - > Satisfaction	0.144	0.148	−0.004	0.462	0.448	No
Comfort & Convenience - > Satisfaction	0.311	0.176	0.135	0.020	0.026	Yes
Connectivity - > Satisfaction	0.290	0.198	0.092	0.037	0.047	Yes
Operations - > Satisfaction	0.209	0.321	−0.112	0.023	0.026	Yes
Safety - > Satisfaction	0.155	0.099	0.056	0.133	0.147	No
Station Facilities - > Satisfaction	0.136	0.250	−0.114	0.030	0.029	Yes
Group 2: (Age ≥40) vs (Age <40)						
Relationship	**Path Coefficient (Age ≥40)**	**Path Coefficient (Age <40)**	**Path Coefficient difference**	**p-value Henseler's MGA**	**Permutation test**	**Significant difference?**
Cleanliness - > Satisfaction	0.206	0.122	0.084	0.035	0.043	Yes
Comfort & Convenience - > Satisfaction	0.233	0.252	−0.019	0.390	0.397	No
Connectivity - > Satisfaction	0.216	0.258	−0.042	0.222	0.255	No
Operations - > Satisfaction	0.240	0.277	−0.037	0.234	0.255	No
Safety - > Satisfaction	0.111	0.135	−0.024	0.312	0.311	No
Station Facilities - > Satisfaction	0.273	0.158	0.115	0.038	0.043	Yes

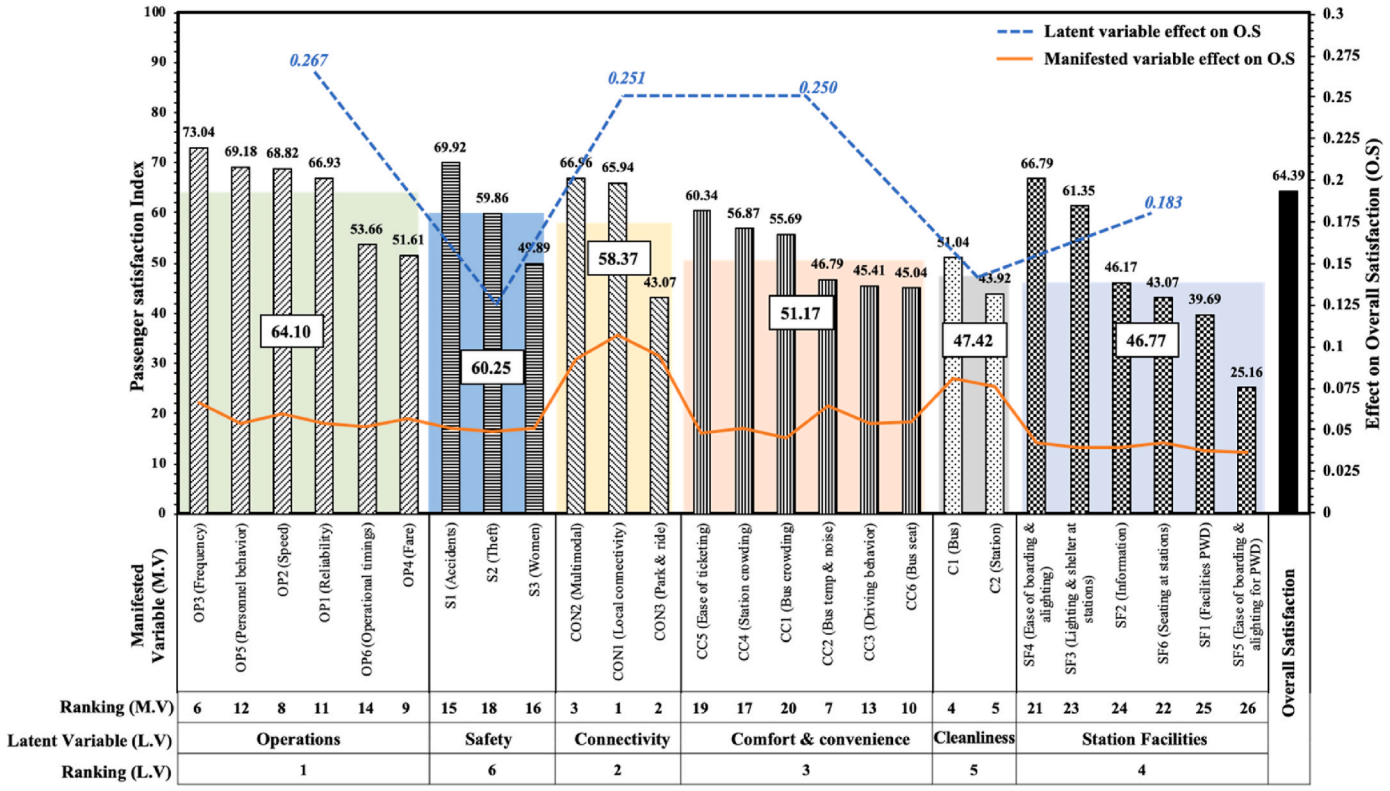


Fig. 5. Passenger satisfaction index and ranking of variables (Overall sample).

passenger satisfaction). Whereas, performance associated with a service dimension was calculated through the performance index (P.I) using Equation (2).

$$P.I = \frac{\sum_{i=1}^{k} \beta_i \overline{y_i} - \sum_{i=1}^{k} \beta_i}{(r-1)\sum_{i=1}^{k} \beta_i} \cdot 100 \quad (2)$$

Where, 'β_i' is the weight of indicator 'i' measured for a service dimension; '$\overline{y_i}$' is the mean value of indicator 'i'; 'k' is the number of indicators in a service dimension, and 'r' is the Likert scale value used in the study. For the performance index, a score lower than 60 signifies very dissatisfied, a score between 60 and 70 signifies relatively dissatisfied, a score between 70 and 80 signifies moderately satisfied, a score between 80 and 90 signifies relatively satisfied, and a score higher than 90 signifies excellent satisfaction (Shen et al., 2016).

As shown in Fig. 5, all the dimensions and indicators had relatively lower performance (performance <80), indicating that the Ahmedabad BRTS needs significant improvement to ensure passengers' overall satisfaction. Among the six dimensions of passenger satisfaction, operations had the highest passenger satisfaction index of 64.10, followed by safety with a satisfaction index of 60.25. Further, the dimension of station facilities had the lowest satisfaction index of 46.77, and the results indicate that the system was severely lacking in providing the essential services required for person with disability (PWD). In terms of ranking the service dimensions based on their importance, the dimension of *"Operations"* had the highest effect on overall passenger satisfaction (ranked 1st), followed by *"Connectivity"* (ranked 2nd), *"Comfort & convenience"* (ranked 3rd), *"Station facilities"* (ranked 4th), *"Cleanliness"* (ranked 5th) and *"Safety"* (ranked 6th). Whereas, in terms of indicators, the three most influential indicators were *"CON1: Local connectivity"* (ranked 1st), *"CON3: Park & ride"* (ranked 2nd), and *"CON2: Multimodal connectivity"* (ranked 3rd). The overall satisfaction index of the Ahmedabad BRTS was not very high i.e., 64.39, and has serious room for improvement to ensure passenger satisfaction, which may improve passengers' ridership.

To identify the dimensions requiring urgent improvement to enhance the overall satisfaction of passengers, Importance performance map analysis (IPMA) was performed in the present study. Fig. 6 (a) presents the results of IPMA for all the respondents participated in the study (n = 710). The graph is divided into four quadrants by calculating the mean value of importance and performance for all the measures. Among all the service dimensions, "*operations*" and "*connectivity*" belonged to the 1st quadrant, where the performance and importance of the dimensions were relatively higher. These were the important dimensions where the BRTS was performing relatively well, and the authorities should only maintain the performance of these dimensions in the future to ensure passenger satisfaction. Quadrant 2 includes the "*safety*" dimension, which was also performing relatively well; however, its importance was less, thereby suggesting that additional efforts in this area can be shifted to other dimensions requiring urgent improvement. The dimension of "*comfort & convenience*" was contained in the 4th quadrant, indicating higher importance and lower performance of the dimension.

In terms of gender and age-specific improvement, Fig. 6 (b, c) shows the results of IMPA for male-female and passengers with age ≥40 - age <40, respectively. For female passengers, it was found that "*comfort & convenience*" lies in the 4th quadrant, thus, requiring higher improvement to improve the overall satisfaction of female passengers. Whereas, for male passengers, the dimension of "*comfort & convenience*" lies in the 3rd quadrant, suggesting no additional efforts are required in this area to improve passenger satisfaction. Moreover, for male passengers, the dimension of "*station facilities*" lies in the 4th quadrant, thereby emphasizing that additional efforts are required in this area to improve the overall satisfaction of male passengers on the Ahmedabad BRTS. Furthermore, for passengers with age ≥40, the dimension of "*comfort & convenience*" and "*station facilities*" lies in the 4th quadrant. However, for passengers with age <40, only the dimension of "*comfort and convenience*" lies in the 4th quadrant, whereas "*station facilities*" dimension

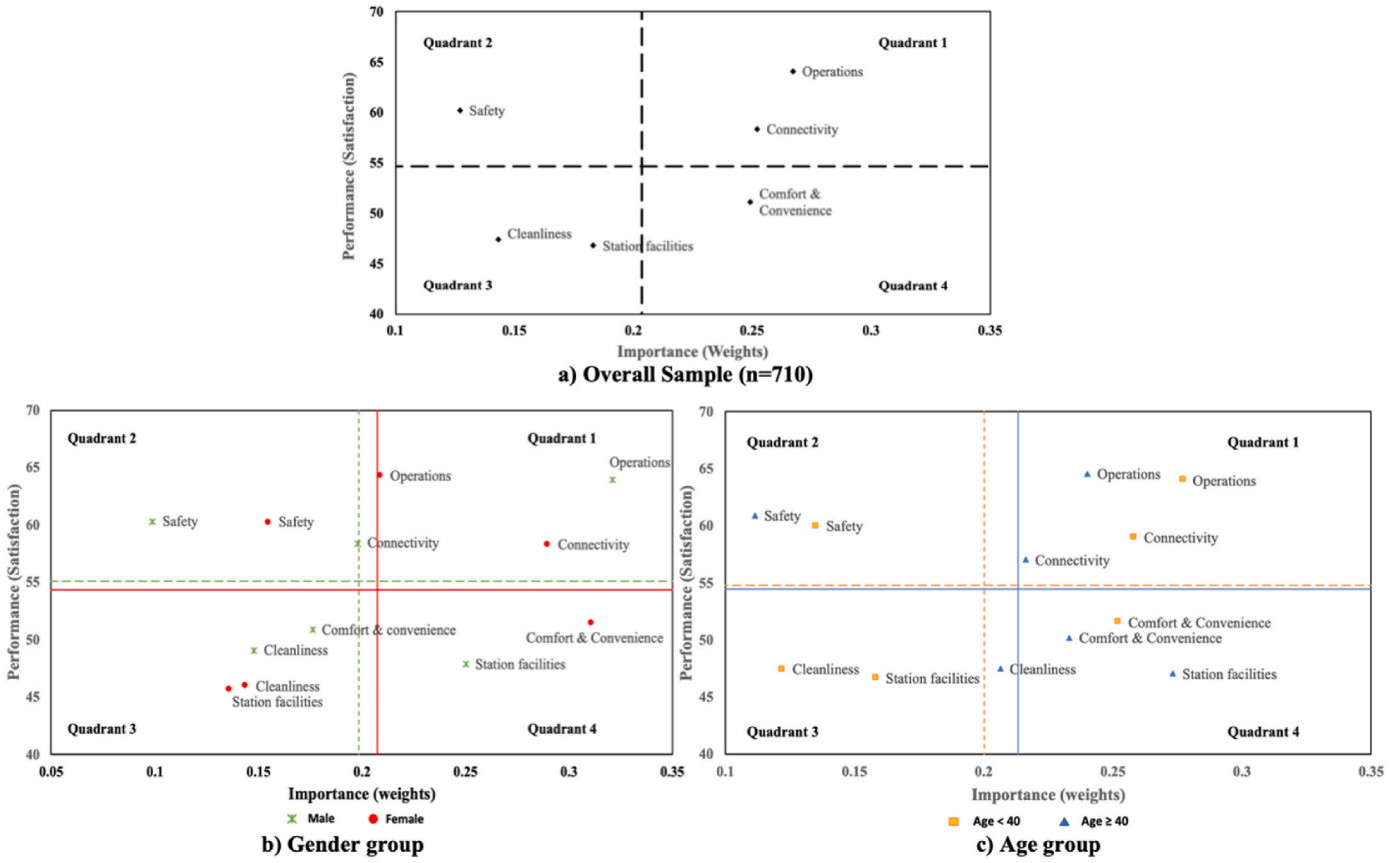


Fig. 6. Importance performance map analysis.

lies in the 3rd quadrant.

Further, the present study performed indicator-level IPMA to identify even more specific areas requiring improvement to enhance the overall passenger satisfaction on the Ahmedabad BRTS. Table 5 reveals that multimodal and local connectivity was an important parameter for different groups of passengers, which was performing relatively well in enhancing the overall satisfaction of passengers. However, in terms of cleanliness (bus and station cleanliness) and park & ride availability near/at the stations, there is an urgent need to put more effort into improving passengers' satisfaction belonging to different groups of age and gender. Further, for the female group of passengers, the results reveals that there is an urgent need to ensure the comfort (bus temp. & noise, driving behavior, and bus seating) and safety of women. Moreover, for passengers with age $\geq$ 40, there is an urgent need to ensure proper seating inside the stations for passengers waiting at the stations for buses.

5. Discussion

5.1. Determinants of passenger satisfaction

The results of PCA revealed six major constructs that explain passenger satisfaction on BRTS, namely *connectivity, comfort & convenience, safety, cleanliness, station facilities and operations*. The results of PLS-SEM strongly support that overall passenger satisfaction is positively associated with these constructs. For the overall sample, the measure of "*operations*" had the highest impact on influencing overall passenger satisfaction. Zhang et al. (2019) and Yuan et al. (2021) also observed similar findings for public transport system in China, where it was observed that "*operations*" had higher influence on overall passenger satisfaction. From the results, it can be inferred that passengers give higher importance to following "*operations*" factor in influencing their overall satisfaction; system's reliability, speed, and frequency of buses, fare, the behavior of BRTS personnel, and operational timing. Allen et al. (2019b) also observed that reliability in terms of speed and frequency had the most significant influence on the satisfaction of passengers on bus transit system in Barcelona. In addition, the present study unveils

Table 5
IPMA indicator level analysis.

Quadrant	Recommendations	Total Sample (n = 710)	Male (n = 345)	Female (n = 365)	Age ≥40 (n = 197)	Age <40 (n = 513)
1	Keep up good work	CON1, CON2, OP2, OP3	CON1, CON2, OP1, OP2, OP3, OP5, SF4	CON1, CON2, S1, S2, CC4, CC5	CON1, CON2, SF4,	CON1, CON2, OP2, OP3, OP5
2	Possible overkill	S1, S2, SF3, SF4, CC1, CC4, CC5, OP1, OP5	S1, S2, SF3, CC1, CC4, CC5	OP1, OP2, OP3, OP5, SF3, SF4, CC1	S1, S2, OP1, OP2, OP3, OP5, CC1, CC4, CC5, SF3	S1, S2, SF3, SF4, OP1, CC1, CC4, CC5
3	No efforts required	SF1, SF2, SF5, SF6, OP4, OP6, S3, CC3, CC6	SF1, SF2, S3, CC2, CC3, CC6, SF5	OP4, OP6, SF1, SF2, SF5, SF6	S3, OP4, OP6, SF1, CC3, SF5, SF2, CC6	OP6, S3, SF1, SF2, SF5, SF6 CC6, CC3
4	Attention required	CON3, C1, C2, CC2,	CON3, C1, C2, OP4, OP6, SF6	CON3, C1, C2, CC2, CC3, CC6, S3	CON3, C1, C2, SF6, CC2	CON3, C1, C2, CC2, OP4

that the *"connectivity"* of the BRTS (in terms of local connectivity and connectivity with multimodal transport) and *"comfort & convenience"* was perceived as an important factor in influencing overall passenger satisfaction. In the past, many studies have also shown that *"connectivity"* and *"comfort & convenience"* are the key determinants in influencing passengers' overall satisfaction (Mandhani et al., 2020; Yuan et al., 2021a; Zhang et al., 2019). The results also highlight that passengers' perceived *"safety"* and *"cleanliness"* to influence their overall satisfaction on the BRTS. However, it was found that passenger perceived "*operations*" and *"comfort & convenience"* higher than "*safety*" and *"cleanliness"*.

5.2. Heterogeneity among passengers groups

Using multigroup analysis, we observed that there exists a disparity among the male and female passengers in terms of their perception of satisfaction on the BRTS. Previous studies also revealed that female and male passengers perceive satisfaction differently (Mandhani et al., 2021; Nguyen-Phuoc et al., 2021). The results show that female passengers perceived "*connectivity*" ($\beta = 0.290$) and "*comfort & convenience*" ($\beta = 0.311$) to have higher influence on their overall satisfaction, whereas male passengers perceived "*operations*" ($\beta = 0.321$) and "*station facilities*" ($\beta = 0.250$) to have higher influence on their overall satisfaction. Higher importance towards *connectivity* can be attributed due to feeling of insecurity and vulnerability to crime among female passengers which would limit their mobility requirement through public transport facilities (ITF, 2018). Females expect the public transport facility to be well connected with all the major locations and paratransit services are available for their trips, providing first and last-mile connectivity so as to ensure their safety. This is also in line with the findings of Arabikhan et al. (2016) and Mandhani et al. (2021), where it was observed that "*connectivity*" was more important for female passengers than male passengers. Further, female passengers are more concerned about their safety in overcrowded areas than male passengers (Mandhani et al., 2021), therefore "*comfort & convenience*" (which included the measures of station crowding, bus crowding and ease of ticketing) has higher importance for female passengers. Moreover, higher importance to *"operations"* among male passengers can be explained by the fact that Ahmedabad has a higher male working population than females (Mahadevia, 2012). Therefore, for male passengers, the operational performance of the BRT system was more important, which included the frequency of buses, speed, reliability, operational timings, and fare. In terms of "*safety*", the results are in agreement with the findings of Mandhani et al. (2021), where it was observed that female passengers perceived safety higher as compared to male passengers while using Metro transit service in India. In terms of *"cleanliness"*, both male and female passengers perceived cleanliness as identical.

For passengers of different age groups, it was observed that passengers with age $\geq$40 and age <40 perceived satisfaction on BRT service distinctly. For passengers with age $\geq$40 "*station facilities*" ($\beta = 0.273$) had the highest influence on overall satisfaction, including the provision of adequate seating facilities at the stations, information signages, ease of boarding and alighting for normal passengers and passengers with disability, facilities for PWD, etc. Whereas passengers with age <40 perceived the measure of *"operations"* ($\beta = 0.277$) to have highest influence on their overall satisfaction, including the reliability of system, speed & frequency of buses, service operational timings, and behavior of BRTS staff. Further, passengers with age $\geq$40 perceived "*cleanliness*" inside the stations and buses higher ($\beta = 0.206$) compared to passengers with age <40 ($\beta = 0.122$). However, in terms of "*comfort & convenience*", *"safety"* and *"connectivity"*, there were no significant differences among the passengers of different age groups **[Table 4]**.

6. Policy implications based on IPMA

The overall satisfaction index of the Ahmedabad BRT system was 64.4, which is far below 74.4 of Select Bus Service (SBS) in New York (Wan et al., 2016). Thereby, indicating that the Ahmedabad BRT system needs significant improvement to ensure passenger satisfaction. The results of the IPMA can be utilized by managers and policymakers of BRTS to identify the dimensions requiring urgent improvement to enhance the overall satisfaction of passengers. Considering the heterogeneity in passenger perception of satisfaction among passengers of different age and gender groups, the present study provides suggestions for BRTS agencies to take appropriate countermeasures to ensure passenger satisfaction for a specific group of passengers **[Fig. 7]**. While it should be noted that the policy implications presented in this study pertain specifically to the IPMA analysis of Ahmedabad BRTS and may not be universally applicable, transit systems facing similar challenges can leverage these recommendations to design and implement effective interventions in both developing and developed countries. The study's central finding of significant differences in satisfaction levels among passengers of different genders and age groups highlights the need to account for these factors in transit satisfaction assessments and develop targeted solutions to effectively address them.

The study findings showed that, for female passengers, "*comfort & convenience*" has the highest influence on overall satisfaction ($\beta = 0.311$). However, its performance was found to be relatively lower, suggesting the need for immediate improvements to enhance female passengers' satisfaction in this area. Statistically, a unit increase in perception of *"comfort & convenience"* can result in 31.1% and 17.6% increase in overall satisfaction for female and male passengers, respectively. Considering, female passengers are more concerned about their safety in overcrowded areas (Mandhani et al., 2021), strategic policies like operating more buses with higher frequency on the corridors with a higher demand and introducing more ticket validation gates at high-ridership stations may help alleviate overcrowding on both buses and stations. Further, indicator-level IPMA revealed that for female passengers, women's safety and cleanliness (bus & station) had high importance with relatively low performance. This underscores the need to prioritize cleanliness and women's safety to ensure female passenger satisfaction on Ahmedabad BRTS. Measures such as installing panic buttons and surveillance cameras on all buses, developing an application to assist women, establishing a dedicated helpline number for women, and training BRTS staff on gender sensitivity and awareness campaigns can be implemented to improve women's safety on the BRTS system.

For male passengers, findings reveal that *"station facilities"* has the second-highest impact on overall satisfaction ($\beta = 0.250$), but its performance was relatively low, indicating the need for urgent improvement. Higher performance for *"station facilities"* can be achieved by improving public information systems (PIS) to provide real-time information to passengers about bus schedules, delays, and expected arrivals. Providing real-time information related to bus delays can reassure passengers that the buses are on their way and accordingly they can utilize their time effectively (TCQSM, 2013). Additionally, passengers who are new to the service or even frequent users require information related to routes, fares, schedule of buses, delays, etc., so that they know how to use the transit service. To ensure this, authorities can i) provide routes map and schedule information at stations and buses, ii) provide visual displays at stations and inside all the buses to help assist passengers with hearing impairment, iii) provide audible announcements at stations and buses to assist passengers with visual impairment, and iv) by providing real-time information through mobile phone application. The study also indicates that adequate night-time lighting and shelter at stations can enhance male passengers' overall satisfaction. It is worth noting that improving *"station facilities"* will have a greater impact on male passengers than female passengers i.e. a unit increase in passengers perception of *"station facilities"*, can result in a 25% and 13.6% increase in overall satisfaction for male and female passengers, respectively.

For passengers with age $\geq$ 40, the results of IPMA graph reveals that the measure of "*station facilities*" and "*comfort & convenience*" require the immediate attention of BRTS authorities to enhance the satisfaction

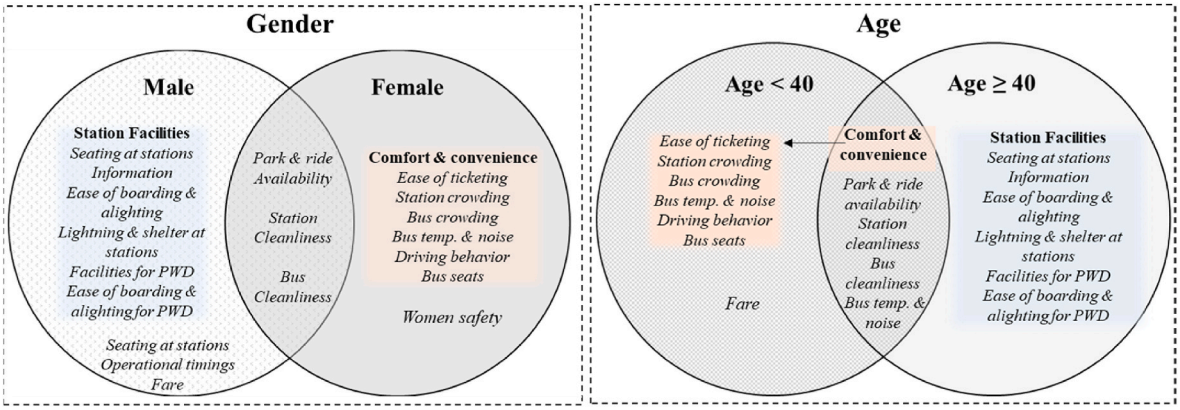


Fig. 7. Areas of improvement for passengers of different age and gender groups based on IPMA.

level of aged passengers (age $\geq$ 40). To achieve this, BRTS authorities can take measures such as i) providing adequate seating facilities at stations to enable aged passengers to sit while waiting for the buses, ii) ensuring easy boarding and alighting for aged passengers, and iii) ease of ticketing for aged passengers. On the other hand, for passengers with age <40, the measure of "*comfort & convenience*" requires greater attention and this can be achieved by i) providing comfortable seating and ensuring that air conditioning is functional in all the buses to ensure comfort inside the buses, ii) organizing driver training programmes aimed at improving the driving behavior of bus drivers to enhance passenger comfort, and iii) promoting the use of intelligent ticketing systems such as use of smart cards and mobile app-based tickets. Considering, that most of the young age passengers were regular commuters, using the service for going to schools, universities and workplaces, it would be beneficial to introduce monthly passes at a lower cost to ensure ridership and satisfaction of passengers.

It is crucial to recognize that implementing interventions tailored to specific passenger groups can have a positive impact on the overall satisfaction of all passengers. For example, improvement in *"station facilities"* will affect all the passengers belonging to different ages and gender. However, improving *"station facilities"* will have a greater impact on overall satisfaction of aged passengers (age $\geq$ 40) ($\beta = 0.273$) compared to passengers with age <40 ($\beta = 0.158$), male passengers ($\beta = 0.250$) and female passengers ($\beta = 0.136$). In developing countries, transit agencies have traditionally only implemented gender and age-neutral interventions in their transit system planning and design. However, this approach does not address the needs of different passenger groups. Therefore a holistic approach focussed on ensuring that transit service meets the required satisfaction levels for passengers of all ages and genders is essential. Furthermore, for low ridership groups, these findings can aid in implementing gender and age-specific measures that will enhance overall satisfaction on the BRTS, leading to an increase in passenger ridership.

7. Conclusion

Given the pressing need to address urban mobility challenges such as traffic congestion and vehicle emissions, and to promote public transport as a sustainable alternative to private vehicles, it is imperative for transit agencies to assess the effectiveness of their current transit systems in ensuring passenger satisfaction. This will allow transit agencies to identify areas for improvement and enhance their services to retain their existing passengers while also attracting new ones to use public transportation. This study, through a comprehensive evaluation methodology, focuses on examining various attributes of BRTS that significantly influence passenger satisfaction and explore the heterogeneity among passengers' perception of satisfaction for passengers belonging to different socioeconomic groups to enhance overall satisfaction on the BRT system. The findings reveals that the attributes that significantly influence the passenger satisfaction on BRTS were *connectivity, comfort & convenience, safety, cleanliness, station facilities and operations.* In addition, the present study examines the less-explored gender and age aspects in bus transit satisfaction evaluation and the findings highlight that there exist disparity in perception of satisfaction for passengers of different age and gender. These findings indicate the need for implementing group-specific interventions to improve passenger satisfaction and suggests tailoring interventions to address the unique needs of passengers based on their demographic variables.

While other sociodemographic and trip-related variables such as trip purpose and frequency, vehicle ownership, education level, and income level are relevant in transit satisfaction evaluation, this study focused only on examining the differences in passenger satisfaction among passengers of different ages and gender groups, given that these are fundamental social identities that shape individuals' perspectives and experiences in various ways. Men and women often have different travel patterns which is primarily due to the distinct safety concerns and societal roles they face (ITF, 2018; Tilley and Houston, 2016). In particular women are more concerned about their safety and have higher household roles and caretaking responsibilities, which may affect their travel behavior and preferences (Mandhani et al., 2021; Tilley and Houston, 2016). Furthermore, travel patterns, preferences, and constrain change over the life course i.e. older adults may have different mobility needs and preferences due to age-related physical limitations or health conditions, while younger passengers, such as students, may have different transportation needs and preferences related to their educational activities. However, as a way forward, researchers could adopt and apply the methodology utilized in this study to comparable transportation systems worldwide, in order to investigate how various sociodemographic and trip-related factors impact passenger satisfaction. Despite the contribution of the present study in the literature, the study's results are geographically limited to BRTS in Ahmedabad, India, and therefore findings may not be generalizable to BRT systems worldwide. Furthermore, the study had a relatively small number of respondents aged over 50 years, as it is challenging for older passengers to respond to online surveys. Therefore, for future studies, it is recommended to conduct face-to-face interviews to increase the participation rate among older passengers.

Authorship statement

All persons who meet authorship criteria are listed as authors, and all authors certify that they have participated sufficiently in the work to take public responsibility for the content, including participation in the concept, design, analysis, writing, or revision of the manuscript. Furthermore, each author certifies that this material or similar material has not been and will not be submitted to or published in any other

publication before its appearance in the Transport Policy Journal.

Authorship contributions

Harpreet Singh: Conceptualization, Methodology, Writing-original draft preparation, Data analysis**Ankit Kathuria:** Supervision, Conceptualization, Methodology, Reviewing and Editing.

Data availability

Data will be made available on request.

Appendix A

Table A1
Discriminant validity: Fornell-Larcker criteria

Latent Variable/Construct	Cleanliness	Comfort & Convenience	Connectivity	Operations	Safety	Satisfaction	Station Facilities
Cleanliness	0.911						
Comfort & Convenience	−0.035	**0.783**					
Connectivity	0.227	0.366	**0.859**				
Operations	0.232	0.439	0.464	**0.775**			
Safety	0.121	0.458	0.429	0.471	**0.845**		
Satisfaction	0.316	0.593	0.647	0.679	0.583	**1.000**	
Station Facilities	0.256	0.449	0.509	0.509	0.495	0.659	**0.777**

Table A2
Discriminant validity: Heterotrait-monotrait ratio of correlation (HTMT)

Latent Variable/Construct	Cleanliness	Comfort & Convenience	Connectivity	Operations	Safety	Satisfaction	Station Facilities
Cleanliness							
Comfort & Convenience	0.065						
Connectivity	0.278	0.423					
Operations	0.281	0.491	0.543				
Safety	0.152	0.542	0.524	0.560			
Satisfaction	0.354	0.627	0.711	0.726	0.651		
Station Facilities	0.303	0.507	0.598	0.581	0.593	0.705	

Table A3
Discriminant validity: Cross loading of observed variables

Observed variable	Latent Variable/Construct					
	Cleanliness	Comfort & Convenience	Connectivity	Operations	Safety	Station Facilities
Bus cleanliness (C1)	**0.917**	−0.047	0.206	0.231	0.117	0.248
Station cleanliness (C2)	**0.905**	−0.016	0.208	0.190	0.104	0.217
Bus crowding (CC1)	−0.066	**0.766**	0.242	0.278	0.326	0.302
Temp. & noise inside bus (CC2)	0.023	**0.822**	0.370	0.436	0.417	0.426
Driving behavior (CC3)	−0.079	**0.799**	0.301	0.321	0.344	0.373
Station crowding (CC4)	−0.057	**0.796**	0.226	0.302	0.324	0.300
Ticketing (CC5)	−0.019	**0.746**	0.243	0.302	0.326	0.316
Bus seat (CC6)	0.013	**0.769**	0.312	0.389	0.395	0.367
Local connectivity (CON1)	0.236	0.355	**0.889**	0.426	0.425	0.493
Multimodal connectivity (CON2)	0.146	0.316	**0.832**	0.379	0.337	0.391
Park & ride availability (CON3)	0.198	0.269	**0.856**	0.389	0.336	0.423
Reliability (OP1)	0.268	0.256	0.353	**0.731**	0.302	0.364
Speed (OP2)	0.170	0.359	0.373	**0.786**	0.410	0.421
Frequency (OP3)	0.129	0.434	0.450	**0.803**	0.451	0.462
Fare (OP4)	0.189	0.328	0.369	**0.794**	0.359	0.377
Personnel Behavior (OP5)	0.176	0.338	0.295	**0.744**	0.322	0.403
Service timings (OP6)	0.157	0.305	0.296	**0.791**	0.327	0.325
Safety against road accidents (S1)	0.120	0.377	0.328	0.403	**0.839**	0.446
Safety against theft (S2)	0.108	0.376	0.35	0.394	**0.854**	0.397
Women safety (S3)	0.079	0.406	0.408	0.397	**0.842**	0.411
Facilities for PWD (SF1)	0.164	0.306	0.430	0.380	0.349	**0.754**
Information (SF2)	0.199	0.351	0.339	0.399	0.410	**0.777**
Lighting & shelter (SF3)	0.288	0.324	0.392	0.398	0.370	**0.747**
Bus boarding & alighting (SF4)	0.211	0.367	0.430	0.424	0.441	**0.799**
Boarding & alighting for PWD (SF5)	0.066	0.337	0.361	0.352	0.391	**0.776**
Seating at stations (SF6)	0.249	0.400	0.420	0.417	0.347	**0.810**

Appendix B

Table B1
Variance inflation factor (VIF) results

Latent variable	Manifested/Observed variable	VIF (Indicator)	VIF (latent variable)
Comfort & Convenience	CC1 (Bus crowding)	1.878	1.520
	CC2 (Bus temp & noise)	1.991	
	CC3 (Driving behavior)	1.973	
	CC4 (Station crowding)	1.989	
	CC5 (Ticketing)	1.711	
	CC6 (Bus seat)	1.736	
Station Facilities	SF1 (Facilities PWD)	1.709	1.777
	SF2 (Information)	1.811	
	SF3 (Lighting & shelter at stations)	1.648	
	SF4 (Boarding)	1.954	
	SF5 (Boarding PWD)	1.888	
	SF6 (Seating at stations)	1.980	
Operations	OP1 (Reliability)	1.646	1.643
	OP2 (Speed)	1.876	
	OP3 (Frequency)	1.933	
	OP4 (Fare)	1.992	
	OP5 (Personnel Behavior)	1.671	
	OP6 (Operational timings)	1.980	
Connectivity	CON1 (Local connectivity)	2.034	1.533
	CON2 (Multimodal)	1.711	
	CON3 (Park & ride)	1.901	
Safety	S1 (Accidents)	1.658	1.574
	S2 (Theft)	1.817	
	S3 (Women)	1.684	
Cleanliness	C1 (Bus)	1.771	1.155
	C2 (Station)	1.771	

Appendix C

Table C1
MICOM Step 2: Composition Invariance

Latent variable	Group 1: Female vs Male			Group 2: (Age ≥40) vs (Age <40)		
	Original Correlation	Confidence Interval	Compositional Invariance?	Original Correlation	Confidence Interval	Compositional Invariance?
Cleanliness	0.999	[0.996; 0.999]	Yes	1.000	[0.994; 0.999]	Yes
Comfort & Convenience	1.000	[0.999; 0.999]	Yes	0.999	[0.998; 0.999]	Yes
Connectivity	1.000	[0.999; 1.000]	Yes	1.000	[0.999; 1.000]	Yes
Operations	1.000	[0.999; 0.999]	Yes	0.999	[0.999; 0.999]	Yes
Safety	0.998	[0.998; 0.999]	Yes	0.999	[0.998; 0.999]	Yes
Satisfaction	1.000	[1.000; 1.000]	Yes	1.000	[1.000; 1.000]	Yes
Station Facilities	1.000	[0.999; 1.000]	Yes	1.000	[0.999; 0.999]	Yes

Table C2
MICOM Step 3: Equal mean and equal variance

Latent variable	Step 3: Part 1: Equal mean assessment			Step 3: Part 2: Equal variance assessment			
	Mean - Original Difference	Confidence Interval	Equal Mean Value?	Variance - Original Difference	Confidence Interval	Equal Variances?	Full measurement Invariance?
Group 1: Female vs Male							
Cleanliness	−0.118	[-0.124; 0.118]	Yes	−0.047	[-0.130; 0.126]	Yes	Yes
Comfort & Convenience	0.038	[-0.121; 0.123]	Yes	0.033	[-0.193; 0.194]	Yes	Yes
Connectivity	−0.013	[-0.125; 0.120]	Yes	−0.005	[-0.163; 0.164]	Yes	Yes
Operations	−0.011	[-0.118; 0.131]	Yes	−0.189	[-0.238; 0.228]	Yes	Yes
Safety	0.036	[-0.118; 0.122]	Yes	0.003	[-0.163; 0.163]	Yes	Yes
Satisfaction	0.022	[-0.126; 0.134]	Yes	−0.062	[-0.210; 0.198]	Yes	Yes
Station Facilities	−0.117	[-0.129; 0.117]	Yes	0.018	[-0.162; 0.163]	Yes	Yes
Group 2: (Age ≥40) vs (Age <40)							
Cleanliness	−0.004	[-0.143; 0.143]	Yes	0.009	[-0.142; 0.126]	Yes	Yes
Comfort & Convenience	−0.060	[-0.138; 0.139]	Yes	−0.023	[-0.222; 0.190]	Yes	Yes
Connectivity	−0.071	[-0.155; 0.134]	Yes	0.076	[-0.197; 0.174]	Yes	Yes
Operations	0.055	[-0.146; 0.138]	Yes	−0.015	[-0.284; 0.246]	Yes	Yes
Safety	0.095	[-0.143; 0.147]	Yes	−0.035	[-0.205; 0.187]	Yes	Yes
Satisfaction	−0.025	[-0.146; 0.137]	Yes	−0.019	[-0.246; 0.213]	Yes	Yes
Station Facilities	0.043	[-0.151; 0.140]	Yes	−0.052	[-0.194; 0.178]	Yes	Yes

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